

1.1. Bazı Özel Matrisler:

MATRİSİN ADI	MATRİSİN ÖZELLİĞİ
Orthogonal Matrix	$A^{-1} = A^T$
Symetric Matrix	$A^T = A$
Skew-Symetric Matrix	$A^T = -A$
Hermitian Matrix	$(\bar{A})^T = A$
Skew-Hermitian Matrix	$-(\bar{A})^T = A$
Idempotent Matrix	$A^2 = A$

Nilpotent Matrix	$A^p = 0$
Impotent Matrix	$A^2 = 0$
Invaluate Matrix	$A^2 = I$
Periodic Matrix	$A^{k+1} = A$ veya $A^k = I$ (k periyod)
Commutative Matrices	$AB = BA$
Anti-Commutative Matrices	$AB = -BA$

MATRİSİN ADI	MATRİSİN ÖZELLİĞİ
Diagonal (Köşegen) Matrix	Köşegen harici tüm elemanları sıfır olan matris
Scalar Matrix	Köşegenler aynı sayı olup, köşegen harici tüm elemanları sıfır olan matris
Zero Matrix	Bütün elemanları sıfır olan matris
Lower Triangular Matrix	Sağ üst tarafı 0 olan matrislerdir
Upper Triangular Matrix	Sol alt tarafı 0 olan matrislerdir.

1.2. Parçalanmış Matris ve Alt Matrisler:

Partitioned matrices:

submatrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 \end{bmatrix}$$

Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\Rightarrow r_1 = [1 \ 2 \ 3], \ r_2 = [4 \ 5 \ 6]$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = [c_1 \ c_2 \ c_3]$$

$$\Rightarrow c_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \ c_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \ c_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

1.3. Matrislerin Özellikleri:

- Eğer A ve B non-singular matrisler ise; $(AB)^{-1} = B^{-1}A^{-1}$ or generally; $(A_1A_2 \dots A_n)^{-1} = A_n^{-1} \dots A_2^{-1}A_1^{-1}$
- $(AB)^T = B^T A^T$
- trA : köşegen üzerindeki elemanların toplamı
- $AC=BC$ and if C is not invertable, then is it possible that $A \neq B$
- Eğer A^{-1} non-singular bir matris ise; $(A^{-1})^{-1} = A$
- Negatif olmayan bir n tam sayısı için; $(A^n)^{-1} = A^{-n} = (A^{-1})^n$

7. Eğer A^T non-singular bir matris ise; $(A^T)^{-1} = (A^{-1})^T$
8. A is an invertible nxn matrix, then $[A : I_n] \approx [I_n : A^{-1}]$
9. $(kA)^{-1} = k^{-1}A^{-1}$

1.4. Matrix Transformation:

$$f(u) = Au$$

1.5. Determinant ve Özellikleri:

1. $A_{ij} = (-1)^{i+j} M_{ij}$
2. $|A^T| = |A|$
3. k is scalar and A is nxn matrix, then $|kA| = k^n |A|$
4. $|A^{-1}| = \frac{1}{|A|}$
5. $A^{-1} = \frac{A^*}{|A|} = \frac{(A^T)^{-1}}{|A|}$
6. $|A| = 0$ ise A^{-1} yoktur. Yani A bir singular matristir.
7. The determinant of a triangular matrix, either upper or lower, is the product of the elements on the main diagonal
8. $|AB| = |A||B|$
9. $A = [r_1 \ r_2 \ r_x]$, $B = [r_1 \ r_2 \ r_y]$ and $C = [r_1 \ r_2 \ r_x + r_y]$ then $|C| = |A| + |B|$. (r_1, r_2, r_x, r_y are rows)

9.1. Denklem Sistemleri: n, değişken sayısı olmak üzere;

		Non-Homogeneous	Homogeneous
$ A \neq 0$ olduğunda; $r_A = r_{Ab} = n$ olur		"Unique Solution"	"Unique Solution" yani "Trivial Solution: (0,0,0)"
$ A = 0$	$r_A = r_{Ab} \neq n$	$(n - r_A)'$ ya bağlı infinite (sonsuz) solution	
	$r_A \neq r_{Ab}$	No solution	

Cramer's Rule: Let A is nxn matrix and $c_1, c_2, \dots, c_n$ is columns of matrix A ($A = [c_1 : c_2 : \dots : c_n]$). If $Ax = b$;

$$x_1 = \frac{|b : c_2 : \dots : c_n|}{|A|}, x_2 = \frac{|c_1 : b : \dots : c_n|}{|A|}, \dots, x_n = \frac{|c_1 : c_2 : \dots : b|}{|A|},$$

9.2. Vektör Uzayı:

$\emptyset \neq V$ bir küme, F bir cisim olsun ve

$$+ : V \times V \rightarrow V : \quad \cdot : F \times V \rightarrow V$$

işlemleriyle tanımlansın. $\theta \in V$ birim olmak üzere aşağıdaki özellikler sağlansın. $\forall x, y, z \in V$ ve

$\forall \alpha, \beta \in F$ için;

- i. $(x + y) + z = x + (y + z)$
- ii. $x + y = y + x$
- iii. $x + \theta = x$
- iv. $x + (-1)x = \theta$
- v. $1.x = x$
- vi. $(\alpha\beta)x = (\alpha\beta)x$
- vii. $\alpha(x + y) = \alpha x + \alpha y$

viii. $(\alpha + \beta) x = \alpha x + \beta x$

Bu durumda V kümesine **F cismi üzerinde bir vektör uzayı** denir. $F = R$ ise reel, $F = C$ ise kompleks vektör uzayı olarak adlandırılır.

1.7. Alt Vektör Uzayı:

V, F cismi üzerinde bir vektör uzayı ve $\emptyset \neq U \subseteq V$ olsun. Eğer $\forall y, y_1, y_2 \in U$ ve $\forall \alpha, \beta \in F$ için;

i. $y_1 + y_2 \in U$

ii. $\alpha y \in U$

veya denk olarak;

i. $\alpha y_1 + \beta y_2 \in U$

koşulları sağlanıyorsa U kümesine, **V in bir alt vektör uzayı** denir.

1.8. Vektörlerin Özellikleri:

VEKTÖRLERİN ADI	ÖZELLİKLERİ
gonal Vectors	$u^t v = 0 \text{ or } v^t u = 0 \text{ or } < u, v \geq 0$

If V is a vector space, then

(a) $0 \odot u = 0$ for any vector u in V .

(b) $c \odot 0 = 0$ for any scalar c .

(c) If $c \odot u = 0$, then either $c = 0$ or $u = 0$.

(d) $(-1) \odot u = -u$ for any vector u in V .

1.9. Span, Basis, Dimensions, Lineer Independence:

Let $v_1, v_2, \dots, v_k$ be vectors in a vector space V . A vector v in V is called a **linear combination** of $v_1, v_2, \dots, v_k$ if

$$v = a_1 v_1 + a_2 v_2 + \dots + a_k v_k = \sum_{j=1}^k a_j v_j$$

for some real numbers $a_1, a_2, \dots, a_k$.

If $S = \{v_1, v_2, \dots, v_k\}$ is a set of vectors in a vector space V , then the set of all vectors in V that are linear combinations of the vectors in S is denoted by

$$\text{span } S \text{ or } \text{span } \{v_1, v_2, \dots, v_k\}.$$

Let S be a set of vectors in a vector space V . If every vector in V is a linear combination of the vectors in S , then the set S is said to **span** V , or V is spanned by the set S ; that is, $\text{span } S = V$.

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in a vector space V are said to be **linearly dependent** if there exist constants $a_1, a_2, \dots, a_k$, not all zero, such that

$$\sum_{j=1}^k a_j \mathbf{v}_j = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}. \quad (1)$$

Otherwise, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are called **linearly independent**. That is, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent if, whenever $a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}$,

$$a_1 = a_2 = \dots = a_k = 0.$$

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, then we also say that the set S is **linearly dependent** or **linearly independent** if the vectors have the corresponding property.

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of n vectors in R^n (R_n). Let A be the matrix whose columns (rows) are the elements of S . Then S is linearly independent if and only if $\det(A) \neq 0$.

■ Basis

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in a vector space V are said to form a **basis** for V if (a) $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ span V and (b) $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent.

Let $V = R^3$. The vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ form a basis for R^3 , called the **natural basis** or **standard basis**, for R^3 . We can readily see how to generalize this to obtain the natural basis for R^n . Similarly, $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$ is the natural basis for R_3 . ■

The natural basis for R^n is denoted by $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{th row};$$

that is, $\mathbf{e}_i$ is an $n \times 1$ matrix with a 1 in the i th row and zeros elsewhere.

The natural basis for R^3 is also often denoted by

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The set of vectors $\{t^n, t^{n-1}, \dots, t, 1\}$ forms a basis for the vector space P_n called the **natural**, or **standard basis**, for P_n .

A vector space V is called **finite-dimensional** if there is a finite subset of V that is a basis for V . If there is no such finite subset of V , then V is called **infinite-dimensional**.

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for a vector space V , then every vector in V can be written in one and only one way as a linear combination of the vectors in S .

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of nonzero vectors in a vector space V and let $W = \text{span } S$. Then some subset of S is a basis for W .

Let $V = R^m$ or R_m and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of nonzero vectors in V . The procedure for finding a subset T of S that is a basis for $W = \text{span } S$ is as follows.

Step 1. Form Equation (1),

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n = \mathbf{0}.$$

Step 2. Construct the augmented matrix associated with the homogeneous system of Equation (1), and transform it to reduced row echelon form.

Step 3. The vectors corresponding to the columns containing the leading 1's form a basis T for $W = \text{span } S$.

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for a vector space V and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r\}$ is a linearly independent set of vectors in V , then $r \leq n$.

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ are bases for a vector space V , then $n = m$.

The **dimension** of a nonzero vector space V is the number of vectors in a basis for V . We often write **dim** V for the dimension of V . We also define the dimension of the trivial vector space $\{\mathbf{0}\}$ to be zero.

The set $S = \{t^2, t, 1\}$ is a basis for P_2 , so $\dim P_2 = 3$.

Let V be the subspace of R_3 spanned by $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix}$. Thus every vector in V is of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3,$$

where a_1 , a_2 , and a_3 are arbitrary real numbers. We find that S is linearly dependent, and $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$ (verify). Thus $S_1 = \{\mathbf{v}_1, \mathbf{v}_2\}$ also spans V . Since S_1 is linearly independent (verify), we conclude that it is a basis for V . Hence $\dim V = 2$. ■

Let V be an n -dimensional vector space.

- (a) If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set of vectors in V , then S is a basis for V .
- (b) If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans V , then S is a basis for V .

Step 1. Solve the given homogeneous system by Gauss–Jordan reduction. If the solution contains no arbitrary constants, then the solution space is $\{\mathbf{0}\}$, which has no basis; the dimension of the solution space is zero.

Step 2. If the solution $\mathbf{x}$ contains arbitrary constants, write $\mathbf{x}$ as a linear combination of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$ with $s_1, s_2, \dots, s_p$ as coefficients:

$$\mathbf{x} = s_1\mathbf{x}_1 + s_2\mathbf{x}_2 + \dots + s_p\mathbf{x}_p.$$

Step 3. The set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\}$ is a basis for the solution space of $A\mathbf{x} = \mathbf{0}$; the dimension of the solution space is p .

1.10. Inner Product:

$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in R^2 , denoted by $\|\mathbf{v}\|$, is by the Pythagorean theorem

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

be two vectors in R^3 . By the law of cosines, we have

$$\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Hence

$$\begin{aligned} \cos \theta &= \frac{\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{v} - \mathbf{u}\|^2}{2\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{(u_1^2 + u_2^2 + u_3^2) + (v_1^2 + v_2^2 + v_3^2)}{2\|\mathbf{u}\| \|\mathbf{v}\|} \\ &\quad - \frac{(v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2}{2\|\mathbf{u}\| \|\mathbf{v}\|} \\ &= \frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{\|\mathbf{u}\| \|\mathbf{v}\|}. \end{aligned}$$

Thus

$$\cos \theta = \frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi. \quad (5)$$

In a similar way, if $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ are nonzero vectors in R^2 and θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then

$$\cos \theta = \frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi. \quad (6)$$

anticipate a generalization of the dot product introduced in the next section.

The **standard inner product**, or **dot product**, on R^2 (R^3) is the function that

assigns to each ordered pair of vectors $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in R^2 ($\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ in R^3) the number $\mathbf{u} \cdot \mathbf{v}$ defined by

$$\begin{aligned} &u_1 v_1 + u_2 v_2 \quad \text{in } R^2 \\ &(u_1 v_1 + u_2 v_2 + u_3 v_3) \quad \text{in } R^3. \end{aligned}$$

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}.$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi.$$

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1.$$

It then follows that two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^2 or R^3 are **orthogonal**, or **perpendicular**, if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

A **unit vector** in R^2 or R^3 is a vector whose length is 1. If $\mathbf{x}$ is any nonzero vector, then the vector

$$\mathbf{u} = \frac{1}{\|\mathbf{x}\|} \mathbf{x}$$

Let V be a real vector space. An **inner product** on V is a function that assigns to each ordered pair of vectors $\mathbf{u}, \mathbf{v}$ in V a real number $(\mathbf{u}, \mathbf{v})$ satisfying the following properties:

- (a) $(\mathbf{u}, \mathbf{u}) \geq 0$; $(\mathbf{u}, \mathbf{u}) = 0$ if and only if $\mathbf{u} = \mathbf{0}_V$
- (b) $(\mathbf{v}, \mathbf{u}) = (\mathbf{u}, \mathbf{v})$ for any $\mathbf{u}, \mathbf{v}$ in V
- (c) $(\mathbf{u} + \mathbf{v}, \mathbf{w}) = (\mathbf{u}, \mathbf{w}) + (\mathbf{v}, \mathbf{w})$ for any $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V
- (d) $(c\mathbf{u}, \mathbf{v}) = c(\mathbf{u}, \mathbf{v})$ for $\mathbf{u}, \mathbf{v}$ in V and c a real scalar

A real vector space that has an inner product defined on it is called an **inner product space**. If the space is finite dimensional, it is called a **Euclidean space**.

Cauchy*-Schwarz† Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are any two vectors in an inner product space V , then

$$|(\mathbf{u}, \mathbf{v})| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Triangle Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are any vectors in an inner product space V , then $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

If V is an inner product space, we define the **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ in V as $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Let V be an inner product space. A set S of vectors in V is called **orthogonal** if any two distinct vectors in S are orthogonal. If, in addition, each vector in S is of unit length, then S is called **orthonormal**.

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a finite orthogonal set of nonzero vectors in an inner product space V . Then S is linearly independent.

Gram–Schmidt† Process

In this section we prove that for every Euclidean space V we can obtain a basis S for V such that S is an orthonormal set; such a basis is called an **orthonormal basis**, and the method we use to obtain it is the **Gram–Schmidt process**. From our work with the natural bases for R^2 , R^3 , and in general, R^n , we know that when these bases are present, the computations are kept to a minimum. The reduction in the computational effort is due to the fact that we are dealing with an orthonormal basis. For example, if $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is a basis for an n -dimensional Euclidean space V , then if $\mathbf{v}$ is any vector in V , we can write $\mathbf{v}$ as

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n.$$

The coefficients $c_1, c_2, \dots, c_n$ are obtained by solving a linear system of n equations in n unknowns.

However, if S is orthonormal, we can produce the same result with much less work. This is the content of the following theorem:

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be an orthonormal basis for a Euclidean space V and let $\mathbf{v}$ be any vector in V . Then

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n,$$

where

$$c_i = (\mathbf{v}, \mathbf{u}_i), \quad i = 1, 2, \dots, n.$$

Gram–Schmidt Process

Let V be an inner product space and $W \neq \{\mathbf{0}\}$ an m -dimensional subspace of V . Then there exists an orthonormal basis $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ for W .

$$a_1 = -a_2 \frac{(\mathbf{u}_2, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)}.$$

We may assign an arbitrary nonzero value to a_2 . Thus, letting $a_2 = 1$, we obtain

$$a_1 = -\frac{(\mathbf{u}_2, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)}.$$

Hence

$$\mathbf{v}_2 = a_1 \mathbf{v}_1 + \mathbf{u}_2 = \mathbf{u}_2 - \frac{(\mathbf{u}_2, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)} \mathbf{v}_1.$$

At this point we have an orthogonal subset $\{\mathbf{v}_1, \mathbf{v}_2\}$ of W (Figure 5.19).

Next, we look for a vector $\mathbf{v}_3$ in the subspace W_2 of W spanned by $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ that is orthogonal to both $\mathbf{v}_1$ and $\mathbf{v}_2$. Of course, W_2 is also the subspace spanned

by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{u}_3\}$ (Why?). Thus $\mathbf{v}_3 = b_1 \mathbf{v}_1 + b_2 \mathbf{v}_2 + b_3 \mathbf{u}_3$. We try to find b_1, b_2 , and b_3 so that $(\mathbf{v}_3, \mathbf{v}_1) = 0$ and $(\mathbf{v}_3, \mathbf{v}_2) = 0$. Now

$$0 = (\mathbf{v}_3, \mathbf{v}_1) = (b_1 \mathbf{v}_1 + b_2 \mathbf{v}_2 + b_3 \mathbf{u}_3, \mathbf{v}_1) = b_1(\mathbf{v}_1, \mathbf{v}_1) + b_3(\mathbf{u}_3, \mathbf{v}_1)$$

$$0 = (\mathbf{v}_3, \mathbf{v}_2) = (b_1 \mathbf{v}_1 + b_2 \mathbf{v}_2 + b_3 \mathbf{u}_3, \mathbf{v}_2) = b_2(\mathbf{v}_2, \mathbf{v}_2) + b_3(\mathbf{u}_3, \mathbf{v}_2).$$

Observe that $\mathbf{v}_2 \neq \mathbf{0}$ (Why?). Solving for b_1 and b_2 , we have

$$b_1 = -b_3 \frac{(\mathbf{u}_3, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)} \quad \text{and} \quad b_2 = -b_3 \frac{(\mathbf{u}_3, \mathbf{v}_2)}{(\mathbf{v}_2, \mathbf{v}_2)}.$$

We may assign an arbitrary nonzero value to b_3 . Thus, letting $b_3 = 1$, we have

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{(\mathbf{u}_3, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)} \mathbf{v}_1 - \frac{(\mathbf{u}_3, \mathbf{v}_2)}{(\mathbf{v}_2, \mathbf{v}_2)} \mathbf{v}_2.$$

We next seek a vector $\mathbf{v}_4$ in the subspace W_3 spanned by $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$, and also by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{u}_4\}$, which is orthogonal to $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. We obtain

$$\mathbf{v}_4 = \mathbf{u}_4 - \frac{(\mathbf{u}_4, \mathbf{v}_1)}{(\mathbf{v}_1, \mathbf{v}_1)} \mathbf{v}_1 - \frac{(\mathbf{u}_4, \mathbf{v}_2)}{(\mathbf{v}_2, \mathbf{v}_2)} \mathbf{v}_2 - \frac{(\mathbf{u}_4, \mathbf{v}_3)}{(\mathbf{v}_3, \mathbf{v}_3)} \mathbf{v}_3.$$

We continue in this manner until we have an orthogonal set

$$T^* = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

of m vectors. By Theorem 5.4 we conclude that T^* is a basis for W . If we now let $\mathbf{w}_i = \frac{1}{\|\mathbf{v}_i\|} \mathbf{v}_i$ for $i = 1, 2, \dots, m$, then $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ is an orthonormal basis for W . ■

Let V be an n -dimensional Euclidean space, and let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be an orthonormal basis for V . If $\mathbf{v} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_n\mathbf{u}_n$ and $\mathbf{w} = b_1\mathbf{u}_1 + b_2\mathbf{u}_2 + \dots + b_n\mathbf{u}_n$, then

$$(\mathbf{v}, \mathbf{w}) = a_1b_1 + a_2b_2 + \dots + a_nb_n.$$

1.11. Linear Transformation

Theorem 6.6 If $L: V \rightarrow W$ is a linear transformation of an n -dimensional vector space V into a vector space W , then

$$\dim \ker L + \dim \text{range } L = \dim V.$$

Let $L: V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W ($n \neq 0, m \neq 0$) and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be ordered bases for V and W , respectively. Then the $m \times n$ matrix A whose j th column is the coordinate vector $[L(\mathbf{v}_j)]_T$ of $L(\mathbf{v}_j)$

with respect to T has the following property:

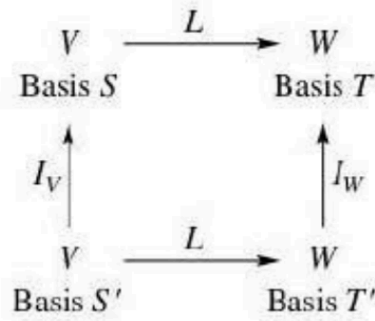
$$[L(\mathbf{x})]_T = A[\mathbf{x}]_S \quad \text{for every } \mathbf{x} \text{ in } V.$$

Moreover, A is the only matrix with this property.

We can summarize the conditions under which a linear transformation L of an n -dimensional vector space V into itself (or, more generally, into an n -dimensional vector space W) is invertible by the following equivalent statements:

1. L is invertible.
2. L is one-to-one.
3. L is onto.
4. The matrix A representing L with respect to ordered bases S and T for V and W is nonsingular.

Let $L: V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W . Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $S' = \{\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_n\}$ be ordered bases for V , with transition matrix P from S' to S ; let $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ and $T' = \{\mathbf{w}'_1, \mathbf{w}'_2, \dots, \mathbf{w}'_m\}$ be ordered bases for W with transition matrix Q from T' to T . If A is the representation of L with respect to S and T , then $Q^{-1}AP$ is the representation of L with respect to S' and T' .



Let $L: V \rightarrow V$ be a linear operator on an n -dimensional vector space. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $S' = \{\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_n\}$ be ordered bases for V with transition matrix P from S' to S . If A is the representation of L with respect to S , then $P^{-1}AP$ is the representation of L with respect to S' . ■

If A and B are $n \times n$ matrices, we say that B is **similar** to A if there is a nonsingular matrix P such that $B = P^{-1}AP$.

Theorem 6.14 Let V be any n -dimensional vector space and let A and B be any $n \times n$ matrices. Then A and B are similar if and only if A and B represent the same linear transformation $L: V \rightarrow V$ with respect to two ordered bases for V . ■

